

- 3 Two progressive water waves X and Y travel along a straight line from point A to point B. The variation of displacement of the waves with distance from A at an instant in time is shown in Fig. 3.1.

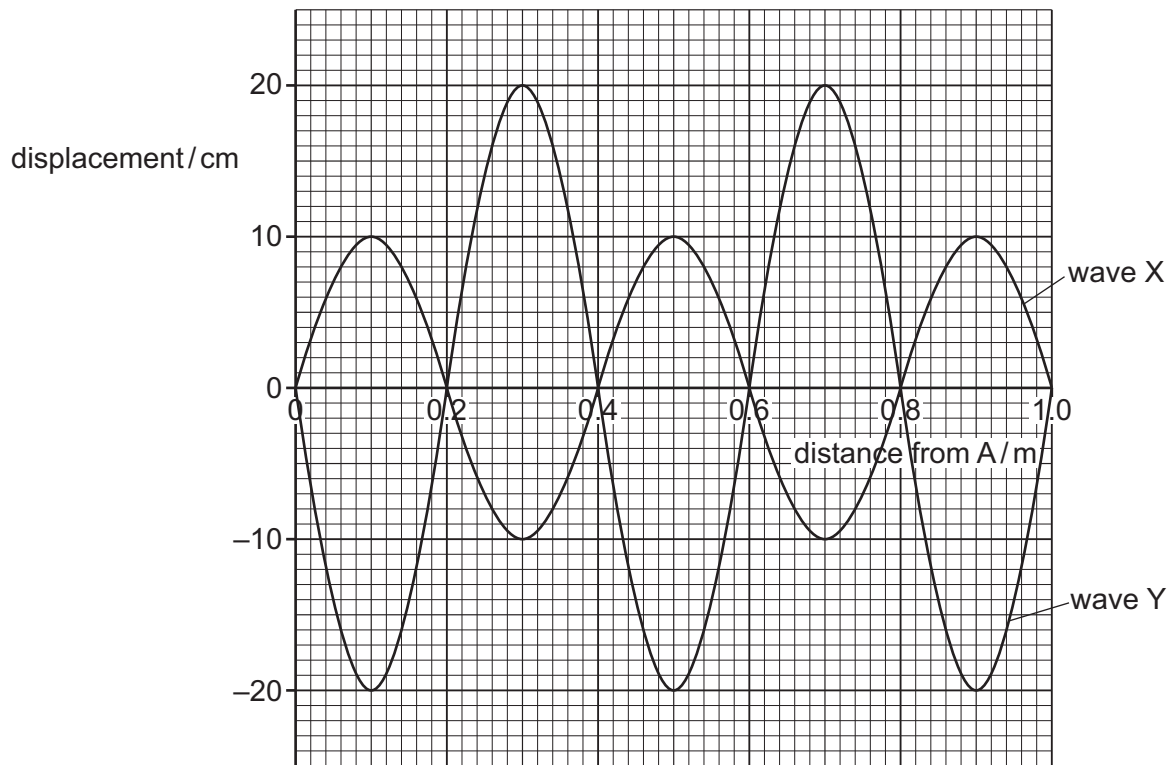


Fig. 3.1

- (a) State the amplitude of wave X.

amplitude = ..... cm [1]

- (b) Both waves have frequency 16 Hz.

- (i) Determine the speed of wave X.

speed = .....  $\text{ms}^{-1}$  [2]

- (ii) State and explain whether X and Y are coherent.

.....  
 .....  
 ..... [1]

